DISTRIBUTED SYSTEMS LABORATORY

PROJECT REPORT

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BACKGROUND

Video streaming traffic makes up a substantial portion of the overall internet traffic and is expected to grow continually. Internet video traffic is expected to grow fourfold between 2017 and 2022, constituting about 82% of the global Internet traffic [1]. In recent years, HTTP-based adaptive streaming has become one of the most predominant forms of video content delivery over the internet. Under this mechanism, the video is divided into chunks, encoded at different bitrates representative of different qualities. Each chunk contains several seconds of video and can be requested at any bitrate, independent of the other chunks. Once received by the NIC, the chunk goes through the video processing pipeline where it is de-multiplexed, parsed, decoded and made ready for rendering. Unplayed chunks are requested and kept in a playback buffer.

With no restriction of downloading the entire video at a single quality, many different ABR algorithms have been proposed. These algorithms use different heuristics to infer the network conditions and choose a bitrate accordingly. The bitrates are chosen to improve certain QoE metrics e.g. maximizing the average bitrate, minimizing rebuffering events, decreasing startup delay etc. These QoE metrics are oftentimes conflicting, in which case, ABR algorithms have to face tradeoffs. The optimality of their decisions then depend on how accurately their chosen tradeoff represents actual users' experience. Moreover, most ABR algorithms make use of heuristics e.g. buffer occupancy, throughput estimates etc. to determine network efficiency. Therefore, there is a lot of scope for improvement to ABR algorithms to ensure maximum user experience.

Some modern ABR algorithms employ a technique called chunk replacement wherein rather than requesting a new video chunk, an existing chunk is requested at a higher quality. In case of available bandwidth, the ABR logic decides between requesting a new chunk or one that is already in the buffer i.e. improving an existing chunk or getting a new chunk with the now available bandwidth. If an existing chunk is requested again, the low quality chunk in the playback buffer is discarded and replaced with the higher

quality chunk in the playback buffer. This allows users to experience improved QoE sooner in the video, at the cost of bandwidth wastage.

KEY IDEAS

Shift to SVC Video Encoding and Chunk Enhancement

Scalable Video Coding (SVC) [5,6] is an extension of the AVC H.264 video compression standard. It involves compression of video chunks such that a single bitstream contains multiple sub-bitstreams. Each of these sub-bitstreams can be independently decoded and rendered to give the same video chunk at a different spatial (frame resolution) or temporal (frames per second) quality.

An SVC Encoded video chunk consists of a Base Layer and several enhancement layers. The base layer (BL) is used to encode the lowest temporal and spatial representations of the video stream, and one or more enhancement layers (ELs) are used to encode additional information. Using the base layer as a starting point, different versions of higher qualities and resolutions can be reconstructed during the decoding process.

Quality Enhancement of SVC Encoded Chunks: Shifting to SVC encoded video chunks as an industry standard will enable more economical quality enhancement of chunks in the playback buffer. With chunk replacement, if the network throughput increases, the ABR logic allows an existing segment to be requested at a higher quality. However, this leads to bandwidth wastage as a single chunk is being requested twice. However, with the adoption of SVC encoding, the entire chunk need not be requested by the replacement logic. By only requesting the relevant enhancement layer and decoding it on top of the already present base layer, we can get an equivalent quality enhancement with fewer bytes fetched. This is expected to prevent bandwidth wastage without significantly compromising the QoE.

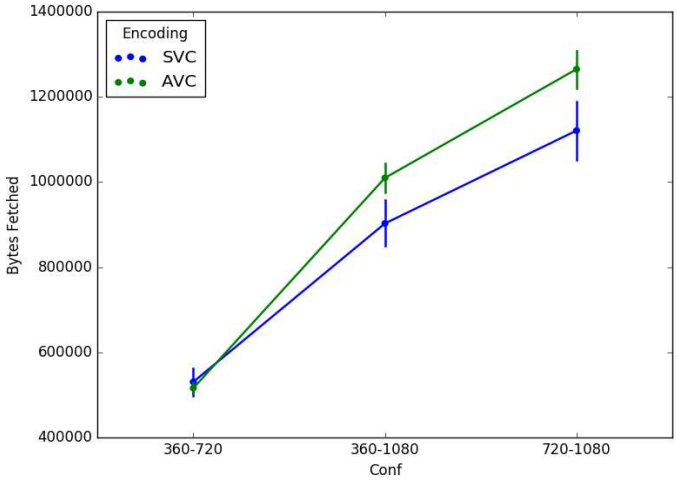
EXPERIMENTAL SETUP

- We implemented quality enhancement over the Sabre [2] simulation tool for ABR algorithms.
- We evaluated our ideas over different baseline ABR algorithms and network traces provided with the Sabre framework.
- We used SVC and AVC chunks of the Big Buck Bunny video [3,4], details of which are as follows:

- Duration: ~10 mins
- 299 segments of 2s each
- Available qualities: 360p, 720p, 1080p
- SVC chunk qualities: 360p (BL), 720p (EL1), 1080p (EL2)

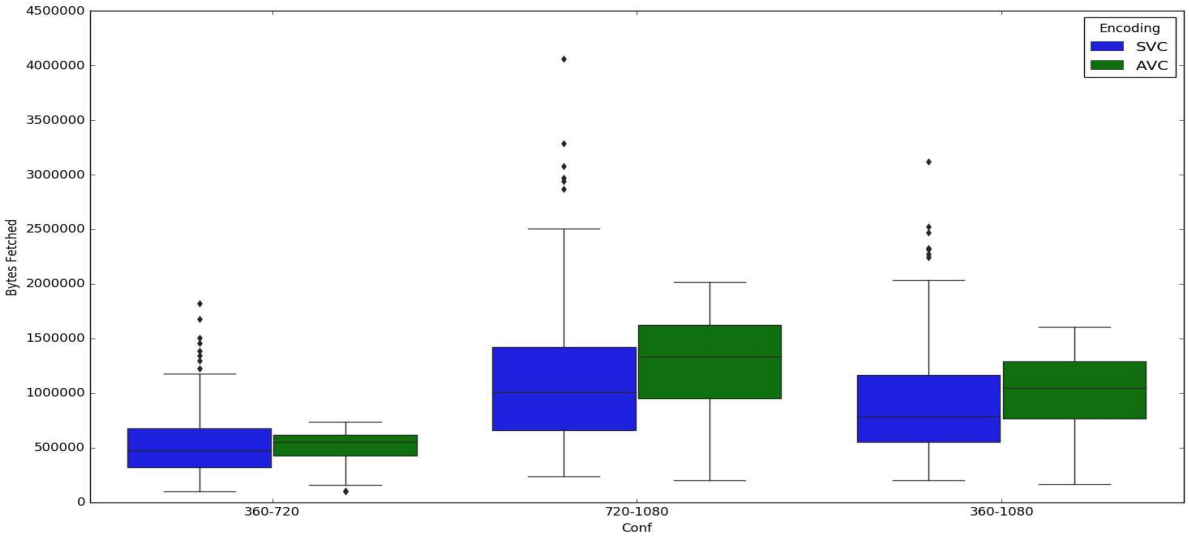
RESULTS AND INSIGHTS

Hypothesis 1: Quality Enhancement using SVC leads to fewer bytes fetched over the network



We compared the average bytes fetched if the quality of a given chunk was improved by replacement vs. if the quality was improved by enhancement.

The plot on the left shows that average bytes fetched are greater for AVC chunks than SVC chunks and this difference becomes more significant as the initial and/or final quality increases.



The average, as well as median bytes fetched is lesser for SVC encoded chunks suggesting that a shift to SVC encoding may help save bandwidth while giving comparable QoE. However, the range of bytes fetched is wider for SVC encoded chunks than AVC with many outliers. There are chunks for which SVC fetches more bytes than AVC in all 3 configurations. This suggests that any possible improvement in QoE will be sensitive to which chunk is being replaced.

Hypothesis 2a: Using SVC encoded videos leads to improved QoE compared to AVC chunks

Using quality enhancement, we expect to gain an improvement in the video quality at a lower cost i.e. with fewer bytes fetched. Therefore, the ABR algorithm should be able to get higher quality chunks more often. This should result in an improved QoE.

Using the Sabre video simulator, and real SVC encoded video chunks, we compared quality enhancement to chunk replacement with different baseline ABR algorithms. We observed that quality enhancement, in fact, leads to lower QoE than chunk replacement. Tables 1 and 2 below summarize our results.

(Time average score)	**Dynamic Dash**	**Throughput**	**Bola**	**Bola-E**
Replacement (AVC)	1.6796	0.8693	1.4643	1.6431
Enhancement (SVC)	1.4255	0.7515	1.4284	1.4047
Enhancement (AVC)	1.7686	0.8653	1.7221	1.7637

(Total Utility)	Dynamic Dash	Throughput	Bola	Bola-E
Replacement (AVC)	502.46	260.06	438.03	491.52
Enhancement (SVC)	426.74	224.78	427.25	422.33
Enhancement (AVC)	529.09	258.85	515.18	527.59

Hypothesis 2b: Loss in QoE of SVC chunks can be attributed to the way SVC chunks are encoded

We changed the distribution of additional bytes that were fetched when doing quality enhancement. Instead of using the SVC chunks/ layers sizes, we simply used the difference in sizes of AVC encoded chunks. For example, for chunk 1, if quality was being enhanced from 720p to 1080p, the size of this EL is the difference in AVC chunk size at these resolutions. We found that this, in fact, does lead to an improvement in the QOE. These results are shown in the last row of Tables 1 and 2.

The observation that AVC enhancement leads to an improvement in QoE while SVC enhancement does not indicates that the loss in QoE can be because of the distribution of SVC chunk sizes. Chunk quality enhancement might prove more beneficial with some other, more efficient compression and coding standards. A more efficient coding standard, where e.g. the distribution of SVC chunk/ layer sizes is closer to the difference in AVC chunks at different qualities, may yield an improvement in QoE.

CONCLUSION

In conclusion, this preliminary study makes a case for considering a shift to different video coding standards (such as SVC) that will enable more economic QoE improvement. The new compression standard, while being efficient, must have a layered structure to prevent wastage of already requested low quality chunks. Such a compression standard will allow us to benefit from improved network conditions sooner, and fewer bytes fetched will allow us to possibly enhance the quality of more chunks in the buffer. Given the massive shift towards use of video

streaming in recent years, research in this direction can allow more economic use of internet bandwidth while improving user QoE.

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